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EXAMINER	
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 18/123,193	Applicant(s) Fang et al.	
	Examiner HAMZA N ALGIBHAH	Art Unit 2441	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 03/17/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-26 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-26 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 03/17/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Detailed Action

1. ***Claims 1-26*** are pending.

2. ***Claims 1-26 are rejected.***

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The filing of a terminal disclaimer by itself is not a complete reply to a nonstatutory double patenting (NSDP) rejection. A complete reply requires that the terminal disclaimer be accompanied by a

reply requesting reconsideration of the prior Office action. Even where the NSDP rejection is provisional the reply must be complete. See MPEP § 804, subsection I.B.1. For a reply to a non-final Office action, see 37 CFR 1.111(a). For a reply to final Office action, see 37 CFR 1.113(c). A request for reconsideration while not provided for in 37 CFR 1.113(c) may be filed after final for consideration. See MPEP §§ 706.07(e) and 714.13.

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The actual filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/apply/applying-online/eterminal-disclaimer.

Claims 14-18 and 20-30 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-20 of U.S. co-pending application No. 18123151. Although the claims at issue are not identical, they are not patentably distinct from each other because the instant claims are broader in scope than the patent claims. It would have been obvious to a person of ordinary skill in the art at the time the invention was made to omit elements wherein the remaining elements perform as before. A person of ordinary skill could have arrived at the present claims by omitting the details of the patented claims. See In re Karlson (CCPA) 136 USPQ 184, decided January 16, 1963 ("Omission of element and its function in combination is obvious expedient if remaining elements perform same functions as before").

This is a provisional nonstatutory double patenting rejection because the patentably indistinct claims have not in fact been patented.

Instant claims	Co-pending claims

1. An apparatus comprising: a parameter acquisition circuit structured to interpret a vehicle parameter value; a property translation circuit structured to interpret a property request value, the property request value defining, at least in part, a requested vehicle property; and a parameter conditioning circuit structured to generate, in response to the property request value, **modified vehicle parameter data** from the vehicle parameter value, **the modified vehicle parameter data** corresponding to the requested vehicle property.	1. An apparatus comprising: a parameter acquisition circuit structured to interpret a vehicle parameter value; a property translation circuit structured to interpret a property request value, the property request value defining, at least in part, a requested vehicle property; a parameter conditioning circuit structured to generate, in response to the property request value, **vehicle property data** from the vehicle parameter value, **the vehicle property data** corresponding to the requested vehicle property; **and** **a vehicle property provisioning circuit structured to transmit the vehicle property data to an external device in response to the property request value.**

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

3. ***Claims 1-4 and 15-18 are rejected*** under 35 U.S.C. 102(a)(2) as being anticipated by Beiderbeck et al (PCT International Application No. PCT/EP2018/060970, published Nov. 22, 2018 wherein the Pub. No.: US 2020/0164882 A1 is used as an English translation for the International Application) hereinafter Beiderbeck.

As per claim 1, Beiderbeck discloses an apparatus, comprising:

- a parameter acquisition circuit structured to interpret a vehicle parameter value (Beiderbeck, abstract, paragraph 0035, 0083, wherein the sensor data 10 and the operating parameters 30 are captured. The operating parameters of the motor vehicle, is captured and the overtaking information is adapted; wherein sensor data and/or the operating parameters can include the vehicle parameter value); and

- a property translation circuit structured to interpret a property request value, the property request value defining, at least in part, a requested vehicle property (Beiderbeck, abstract, paragraph 0022, 0023, wherein there is provision for the overtaking information to be produced on the basis of training data, with at least one of the following properties that characterize the driver. The training data thus have at least one of the properties. As such, the property can be available for example as a speed of travel relative to the mandatory speed, as a safety distance when pulling back in after an effected overtaking maneuver or as a distance from the target vehicle when pulling out before an effected overtaking maneuver. Thus, for example, the overtaking information including at least one of the property of the speed of travel, the safety distance and the distance from the target vehicle can be the property request values as claimed); ***and***

- ***a parameter conditioning circuit structured to generate, in response to the property request value, modified vehicle parameter data from the vehicle parameter value, the***

modified vehicle parameter data corresponding to the requested vehicle property {Beiderbeck, abstract, paragraph 0028-0029, 0031, there is preferably provision for a current operating parameter of the motor vehicle to be captured and for the operating parameter and the overtaking information to be used to recognize an intention of the driver to overtake, and for the control signal to be selected on the basis of the recognized intention to overtake. The operating parameter can be concordant with the properties of the training data, for example. As such, for example the operating parameter allows a determined distance behavior or a determined acceleration behavior to be captured that has a level of similarity with the overtaking information. To this end, the operating parameter can be compared with the overtaking information and then for example categorized as similar or dissimilar on the basis of the comparison. If the operating parameter is assessed as similar to the overtaking information, this then allows the inference that the driver intends to overtake, for example. On the basis of the recognized intention to overtake, for example the control signal allows advice to be output to the driver indicating that the overtaking maneuver would be possible with less risk at a specific distance or else that the present overtaking opportunity should be taken because there is no overtaking opportunity at all in the next journey section, wherein the modified vehicle parameter data can be the outputted control signal resulting from processing of the vehicle properties included within the operating parameters and overtaking information}***;***

As pre claim 2, claim 1 is incorporated and Beiderbeck discloses **wherein the modified vehicle parameter data includes a virtual vehicle property value** (Beiderbeck, abstract, paragraph 0028-0029, 0031, wherein the outputted control signal is virtual since it is outputted as a result of processing (calculating/comparing) the operating parameters and overtaking information);

As pre claim 3, claim 2 is incorporated and Beiderbeck discloses wherein the virtual vehicle property value is based at least in part on two or more vehicle parameter values (Beiderbeck, abstract, paragraph 0028-0029, 0031, wherein the operating parameter can be concordant with the properties of the training data, for example. As such, for example the operating parameter allows a determined distance behavior or a determined acceleration behavior to be captured that has a level of similarity with the overtaking information. To this end, the operating parameter can be compared with the overtaking information and then for example categorized as similar or dissimilar on the basis of the comparison. If the operating parameter is assessed as similar to the overtaking information, this then allows the inference that the driver intends to overtake, for example. On the basis of the recognized intention to overtake, for example the control signal allows advice to be output to the driver indicating that the overtaking maneuver would be possible with less risk at a specific distance or else that the present overtaking opportunity should be taken because there is no overtaking opportunity at all in the next journey section, wherein the modified vehicle parameter data can be the outputted control signal resulting from processing of the vehicle properties included within the operating parameters and overtaking information);

As pre claim 4, claim 2 is incorporated and Beiderbeck discloses wherein the virtual vehicle property value is based at least in part on one or more vehicle parameter values (Beiderbeck, abstract, paragraph 0028-0029, 0031, wherein the operating parameter can be concordant with the properties of the training data, for example. As such, for example the operating parameter allows a determined distance behavior or a determined acceleration behavior to be captured that has a level of similarity with the overtaking information. To this

end, the operating parameter can be compared with the overtaking information and then for example categorized as similar or dissimilar on the basis of the comparison. If the operating parameter is assessed as similar to the overtaking information, this then allows the inference that the driver intends to overtake, for example. On the basis of the recognized intention to overtake, for example the control signal allows advice to be output to the driver indicating that the overtaking maneuver would be possible with less risk at a specific distance or else that the present overtaking opportunity should be taken because there is no overtaking opportunity at all in the next journey section, wherein the modified vehicle parameter data can be the outputted control signal resulting from processing of the vehicle properties included within the operating parameters and overtaking information};

Claims 15-18 are rejected under the same rationale as claims 1-4.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

4. ***Claim***(s) **5-10 and 19-22** is/are rejected under 35 U.S.C. 103 as being unpatentable over Beiderbeck et al (PCT International Application No. PCT/EP2018/060970, published Nov. 22, 2018 wherein the Pub. No.: US 2020/0164882 A1 is used as an English translation for the International Application) in view of Petousis et al (Pub. No.: US 2018/0261020 A1).

As pre claim 5, claim 1 is incorporated and Beiderbeck does not explicitly disclose **wherein the parameter conditioning circuit comprises: a formatting circuit structured to format the vehicle parameter value to a desired format of the requested vehicle property such that the modified vehicle parameter data has the desired format.** However, converting data to a different format is well known in the art. For example, Petousis discloses **wherein the parameter conditioning circuit comprises: a formatting circuit structured to format the vehicle parameter value to a desired format of the requested vehicle property such that the modified vehicle parameter data has the desired format** (Petousis, paragraph 0062 wherein transforming vehicle sensor data into message data functions to convert vehicle sensor data into a suitable format for storage and/or transmission, after receipt of the vehicle sensor data from the scheduling module. Message data shall be understood to be data that is in a suitable format for storage and/or transmission (e.g., a packet, bitstream, buffered message, etc.) Transforming vehicle sensor data into message data can additionally or alternatively be performed in any suitable manner).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify Beiderbeck in view of Petousis such that a formatting circuit is used as claimed because this would have provided a way to ensure that collected and/or processed data can be stored and transmitted using the format needed or more suitable for the storing and/or transmission which improves the efficiency of the system and help reduce real time data loss and degradation (see Petousis paragraph 0003).

As pre claim 6, claim 5 is incorporated and Petousis further discloses **wherein the desired format comprises converting to desired units for the vehicle parameter value** (Petousis, paragraph 0062 wherein transforming vehicle sensor data into message data functions to convert vehicle sensor data into a suitable format for storage and/or transmission, after receipt of the vehicle sensor data from the scheduling module. Message data shall be understood to be data that is in a suitable format for storage and/or transmission (e.g., a packet, bitstream, buffered message, etc.) Transforming vehicle sensor data into message data can additionally or alternatively be performed in any suitable manner).

As pre claim 7, claim 5 is incorporated and Petousis further discloses **wherein the desired format is based at least in part on a network protocol** (Petousis, paragraph 0062 wherein transforming vehicle sensor data into message data functions to convert vehicle sensor data into a suitable format for storage and/or transmission, after receipt of the vehicle sensor data from the scheduling module. Message data shall be understood to be data that is in a suitable format for storage and/or transmission (e.g., a packet, bitstream, buffered message, etc.) Transforming vehicle sensor data into message data can additionally or alternatively be performed in any suitable manner).

As pre claim 8, claim 5 is incorporated and Petousis further discloses **wherein the desired format is based at least in part on storage of the vehicle parameter value in a non-transitory computer readable medium** (Petousis, paragraph 0062 wherein transforming vehicle sensor data into message data functions to convert vehicle sensor data into a suitable format for storage and/or transmission, after receipt of the vehicle sensor data from the scheduling module. Message data shall be understood to be data that is in a suitable format for storage

and/or transmission (e.g., a packet, bitstream, buffered message, etc.) Transforming vehicle sensor data into message data can additionally or alternatively be performed in any suitable manner).

As pre claim 9, claim 8 is incorporated and Petousis further discloses **wherein the desired format is based at least in part on a compression standard** (Petousis, paragraph 0062 wherein transforming vehicle sensor data into message data functions to convert vehicle sensor data into a suitable format for storage and/or transmission, after receipt of the vehicle sensor data from the scheduling module. Message data shall be understood to be data that is in a suitable format for storage and/or transmission (e.g., a packet, bitstream, buffered message, etc.) Transforming vehicle sensor data into message data can additionally or alternatively be performed in any suitable manner).

As pre claim 10, claim 8 is incorporated and Petousis further discloses **wherein the desired format is based at least in part on an encryption scheme** (Petousis, paragraph 0062 wherein transforming vehicle sensor data into message data functions to convert vehicle sensor data into a suitable format for storage and/or transmission, after receipt of the vehicle sensor data from the scheduling module. Message data shall be understood to be data that is in a suitable format for storage and/or transmission (e.g., a packet, bitstream, buffered message, etc.) Transforming vehicle sensor data into message data can additionally or alternatively be performed in any suitable manner).

Claims 19-22 are rejected under the same rationale as claims 5-10.

5. ***Claim***(s) **11 and 23** is/are rejected under 35 U.S.C. 103 as being unpatentable over Beiderbeck et al (PCT International Application No. PCT/EP2018/060970, published Nov. 22, 2018 wherein the Pub. No.: US 2020/0164882 A1 is used as an English translation for the International Application) in view of Renac et al (Pub. No.: US 2015/0244806 A1).

As pre claim 11, claim 1 is incorporated and Beiderbeck does not explicitly disclose wherein the parameter conditioning circuit comprises: a unit conversion circuit structured to convert one or more units of the vehicle parameter value to one or more desired units of the requested vehicle property such that the modified vehicle parameter data has the desired one or more units. However, converting units is well known in the art. For example, Renac discloses wherein the parameter conditioning circuit comprises: a unit conversion circuit structured to convert one or more units of the vehicle parameter value to one or more desired units of the requested vehicle property such that the modified vehicle parameter data has the desired one or more units (Renac, paragraph 0121-0122 wherein a unit conversion factor for restoring the useful datum in a single and homogeneous unit; for example, a conversion factor is applied to the useful datum associated with the displacement speed of the vehicle for expressing it in kilometers/h and not in meters/s or in miles/h, a conversion factor is applied to the useful datum associated with the consumption of the vehicle for expressing it in liters/100 km and not in mm.sup.3/s or .mu.l/s etc).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify Beiderbeck in view of Renac such that a unit conversion circuit is used as claimed because this would have provided a way for extracting data giving the possibility of facilitating development of generic software solutions intended for

machines including a communication bus, while limiting the amount of extracted data from this bus (see BAI paragraph 0019-0022).

Claim 23 is rejected under the same rationale as claim 11.

6. ***Claim***(s) **12-14 and 24-26** is/are rejected under 35 U.S.C. 103 as being unpatentable over Beiderbeck et al (PCT International Application No. PCT/EP2018/060970, published Nov. 22, 2018 wherein the Pub. No.: US 2020/0164882 A1 is used as an English translation for the International Application) in view of BAI et al (Pub. No.: US 2016/0328197 A1).

As pre claim 12, claim 1 is incorporated and Beiderbeck does not explicitly disclose wherein the parameter conditioning circuit comprises: a sampling circuit structured to adjust a sampling rate of the vehicle parameter value to a desired sampling rate of the requested vehicle property such that the modified vehicle parameter data has the desired sampling rate. However, adjusting sampling rate is well known in the art. For example, BAI discloses wherein the parameter conditioning circuit comprises: a unit conversion circuit structured to convert one or more units of the vehicle parameter value to one or more desired units of the requested vehicle property such that the modified vehicle parameter data has the desired one or more units (BAI, paragraph 0022, wherein the QoS module decision module 32 determines a quality level that will be provided for a specific data flow to the portable device 10, such as high-grade quality data, medium-grade quality data, or low-grade quality data. The QoS decision module implements the above functionality utilizing two tuning parameters: a first tuning parameter is a resolution tuning parameter and a second tuning parameter is a sampling rate

parameter. The resolution tuning parameter can be adjusted to obtain a respective degree of resolution to achieve the targeted level of quality. In addition, the sampling rate parameter may be adjusted to sample at a rate for obtaining the desired quality level).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify Beiderbeck in view of BAI such that a sampling circuit is used as claimed because this would have provided a way to improve the efficiency of the system by preventing the involving of excess data (in terms of accuracy and/or frequency) not actually needed and may burden the resource capabilities of the vehicle in providing the data (see BAI paragraph 0022).

As pre claim 13, claim 12 is incorporated and BAI discloses **wherein the sampling circuit is structured to up-sample the vehicle parameter value** (BAI, paragraph 0022, wherein the QoS module decision module 32 determines a quality level that will be provided for a specific data flow to the portable device 10, such as high-grade quality data, medium-grade quality data, or low-grade quality data. The QoS decision module implements the above functionality utilizing two tuning parameters: a first tuning parameter is a resolution tuning parameter and a second tuning parameter is a sampling rate parameter. The resolution tuning parameter can be adjusted to obtain a respective degree of resolution to achieve the targeted level of quality. In addition, the sampling rate parameter may be adjusted to sample at a rate for obtaining the desired quality level);

As pre claim 14, claim 12 is incorporated and BAI discloses **wherein the sampling circuit is structured to down-sample the vehicle parameter value** (BAI, paragraph 0022, wherein the QoS module decision module 32 determines a quality level that will be provided for a specific

data flow to the portable device 10, such as high-grade quality data, medium-grade quality data, or low-grade quality data. The QoS decision module implements the above functionality utilizing two tuning parameters: a first tuning parameter is a resolution tuning parameter and a second tuning parameter is a sampling rate parameter. The resolution tuning parameter can be adjusted to obtain a respective degree of resolution to achieve the targeted level of quality. In addition, the sampling rate parameter may be adjusted to sample at a rate for obtaining the desired quality level);

Claims 24-26 are rejected under the same rationale as claims 21-14.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HAMZA N ALGIBHAH whose telephone number is (571)270-7212. The examiner can normally be reached 7:30 am - 3:30 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Wing Chan can be reached on (571) 272-7493. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/HAMZA N ALGIBHAH/
Primary Examiner, Art Unit 2441